

Gesture Recognition Through Mechanomyogram Signals: An Adaptive Framework for Arm Posture Variability

Panipat Wattanasiri, Samuel Wilson, Weiguang Huo , and Ravi Vaidyanathan 

Abstract—In hand gesture recognition, classifying gestures across multiple arm postures is challenging due to the dynamic nature of muscle fibers and the need to capture muscle activity through electrical connections with the skin. This paper presents a gesture recognition architecture addressing the arm posture challenges using an unsupervised domain adaptation technique and a wearable mechanomyogram (MMG) device that does not require electrical contact with the skin. To deal with the transient characteristics of muscle activities caused by changing arm posture, Continuous Wavelet Transform (CWT) combined with Domain-Adversarial Convolutional Neural Networks (DACNN) were used to extract MMG features and classify hand gestures. DACNN was compared with supervised trained classifiers and shown to achieve consistent improvement in classification accuracies over multiple arm postures. With less than 5 minutes of setup time to record 20 examples per gesture in each arm posture, the developed method achieved an average prediction accuracy of 87.43% for classifying 5 hand gestures in the same arm posture and 64.29% across 10 different arm postures. When further expanding the MMG segmentation window from 200 ms to 600 ms to extract greater discriminatory information at the expense of longer response time, the intra-posture and inter-posture accuracies increased to 92.32% and 71.75%. The findings demonstrate the capability of the proposed method to improve generalization throughout dynamic changes caused by arm postures during non-laboratory usages and the potential of MMG to be an alternative sensor with comparable performance to the widely used electromyogram (EMG) gesture recognition systems.

Index Terms—Arm posture, continuous wavelet transform, domain adaptation, domain adversarial, hand gesture recognition, mechanomyogram.

I. INTRODUCTION

HAND gesture recognition is a prominent research topic for developments of intuitive and automated human-machine interfaces, including medical devices [1], [2], rehabilitation devices [3], and industrial robots [4], [5]. Although the last decades have seen consistent advancement in hand gesture recognition research, existing gesture recognition systems still struggle to achieve satisfactory performance in non-laboratory environments. Dynamic factors in non-laboratory environments such as muscle fatigue [6], [7], positional sensor shift [8], [9], [10], and upper limb position [11], [12], [13] are major obstructions that cause deteriorating performance for gesture recognition, compared to that in controlled environments.

Electromyogram (EMG), widely used in hand gesture recognition research as a data acquisition method, has achieved high prediction accuracy in controlled environments [14], [15]. However, significant decreases in prediction accuracy were presented in EMG-based research when used in non-laboratory environments. For example, up to 27% accuracy drop was presented in long-term use over multiple days with seven gestures classification [16], [17]. Meanwhile, an average drop of 14% was shown in usage over multiple arm postures with the use of time-domain features and a Linear Discriminant Analysis (LDA) classifier [18]. A marked disadvantage of the EMG is the susceptibility to changes in skin impedance caused by perspiration [19], [20], which is a major problem when used in non-laboratory environments. Moreover, commercial EMG interfaces normally require significant preparation time and frequent calibration routines to ensure a relatively high prediction accuracy across different tasks (i.e., different arm postures).

Alternatively, mechanomyogram (MMG), which records the mechanical vibration caused by muscle activities, is an emerging sensing modality and has recently shown comparable performance to the mature EMG systems [21], [22]. Unlike EMG, MMG is less susceptible to changes in skin impedance and requires no specific preparation of the skin [23], [24]. With the comparable performance to EMG and the robustness to impedance changes, MMG has been considered a promising alternative sensor for human-machine interface applications in

Received 4 December 2023; revised 21 June 2024; accepted 11 October 2024. Date of publication 28 October 2024; date of current version 7 April 2025. This work was supported in part by the U.K. EPSRC under Grant EP/R511547/1, in part by the U.K. DRI-CRT, in part by the U.K. NIHR under Grant NIHR202133, and in part by the SERG Technologies Ltd., U.K. (Corresponding authors: Weiguang Huo; Ravi Vaidyanathan.)

This work involved human subjects or animals in its research. Approval of all ethical and experimental procedures and protocols was granted by Imperial College Research Ethics Committee under Application No. 15IC3068.

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Digital Object Identifier 10.1109/JBHI.2024.3483428

real-world environments. However, previous works on MMG-based gesture recognition research mainly focused on applications in controlled environments; only a few studies had aimed to address performance reduction issues caused by changes in muscle characteristics during arm movements.

This paper proposes a new MMG gesture recognition method that aims to address the issue caused by changes in arm postures, which is a major challenge in non-laboratory usage. After training with labeled samples from an initial arm posture (source domain), the proposed classification method can perform unsupervised learning with unlabeled samples collected in a new arm posture (target domain) to improve its generalization during usage. The use of this domain adaptation approach reflects a real-world scenario in which users have a short training time in specific arm postures. During usage, the arm posture can arbitrarily change without the new labeled data provided for the classifier to recalibrate itself.

This work makes several contributions to the field of hand gesture recognition and mechanomyogram (MMG) study as follows:

- A novel MMG-tailored classification pipeline for accurate hand gesture classification in real-time environments with adaptability to dynamic changes from arm postures, which is a common problem encountered in non-laboratory settings.
- A novel use of Continuous Wavelet Transform (CWT) features in combination with Convolutional Neural Network (CNN) models and the domain-adversarial training technique to extract hand gesture information from MMG signals. This novel combination, not previously implemented in gesture recognition literature, enables the pipeline to adapt to new arm postures effectively with the use of unlabeled data from the new domains.
- An investigation on the consistency of MMG signals from the upper limb across multiple arm postures and the impacts caused by key joint movements such as elbow, shoulder, and wrist rotation.

To the best of our knowledge, this is the first study to explore the combination of a domain adaptation technique with mechanomyogram sensing modality to address the robustness problem in hand gesture recognition caused by arm posture changes in non-laboratory environments.

II. RELATED WORKS

For most existing hand gesture recognition systems, the general operation is to train a model with labeled data in an initial setting before using the trained model to predict a newly generated sample. During usage, real-world factors such as fatigue, sensor shifts, or arm movements could cause new samples to diverge from the initial training session and reduce prediction accuracy. Although this problem could be alleviated by frequently calibrating the model with new labeled samples, it could be time-consuming and is not always feasible for practical usage. Conversely, unlabeled samples passively generated during usage are usually not utilized, whereas they are also a valuable source of information that could be leveraged with domain adaptation

techniques to improve accuracy without requiring dedicated training time from users.

Domain adaptation is a technique that aims to address the information changes between datasets and allows the model to perform similar tasks through different datasets. After learning with the training set (source domain), the model can adapt to a new dataset (target domain), which has different characteristics but still retains a similar distribution (e.g., similar class labels). A classic example of domain adaptation applications is classifying color digits images after training the model with grayscale images [25]. The benefits of domain adaptation are especially significant when implementing a model in non-laboratory applications with significant variation of target datasets.

Various studies have explored the use of domain adaptation in hand gesture recognition. Rahimian et al. [26] proposed a transfer learning approach to adapt classifiers to EMG signals from different sessions, subjects, or gestures using few-shot observation from the target domains. By iterative training with small sets of labeled samples from the source domain, the model learns to generalize with only a small portion of labeled data. The model then can achieve generalization through supervised learning on a few samples available in a new domain.

Alternative to supervised learning, gesture recognition applications can be framed as unsupervised domain adaptation problems. For this kind of problem, only labeled datasets from the initial session are available for training, whereas the classifier has to adapt to unlabeled datasets received during usages, which could come from a different domain. Among unsupervised domain adaptation techniques developed in the deep learning field, Domain-Adversarial Training of Neural Networks (DANN) is a popular approach due to its effectiveness, simplicity, and usability [25]. DANN aims to achieve generalization among classes while simultaneously penalizing the model's capability to differentiate between domains. Resulting in a model that focuses on features with good class separability but is indiscriminate to the shifts between domains.

A few studies have introduced the domain-adversarial training concept into their hand gesture recognition systems. Côté-Allard et al. [27] proposed a Self-Calibrating Asynchronous Domain Adversarial Neural Network (SCADANN) to adapt the model to EMG datasets recorded over three weeks. A CNN model was used with spectrogram features to perform unsupervised learning while calibrating itself with a pseudo-labeling scheme. Furthermore, the same group proposed a new model called Transferable Adaptive Domain Adversarial Neural Network (TADANN), which additionally leverages transfer learning into the recalibration process for inter-session gesture recognition [28].

While the above domain adaptation works significantly advance the gesture recognition field, the implementation was tailored toward EMG signals, which inherently have different characteristics from MMG. For example, microphone-based MMG recorded from muscle contractions has transient characteristics compared to EMG, making it infeasible to perform sliding window extraction over the entire contraction to obtain multiple examples as done in EMG applications. As a result, to explore MMG potentials, different feature sets and

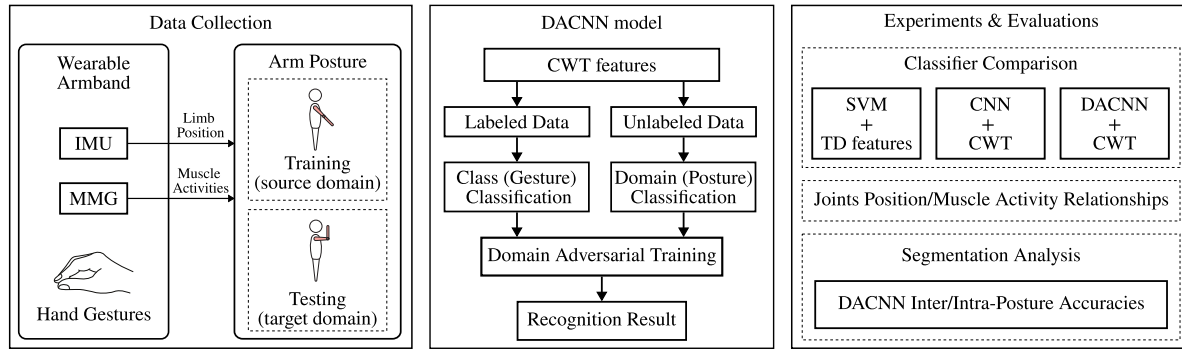


Fig. 1. Diagram of the proposed MMG-based hand gesture recognition architecture and the findings of this work.

classification pipelines specifically designed to extract discriminatory information from MMG signals are necessary.

Inspired by the domain-adversarial training technique used in image recognition and EMG gesture recognition applications [25], [27], [28], this study aims to explore the use of this unsupervised domain adaptation technique on MMG feature spaces and address the problem of arm posture variability, which is a major dynamic factor impacting performances in non-laboratory usage. To extract discriminatory features suitable for the domain adaptation, Continuous Wavelet Transform (CWT) was used with a Convolutional Neural Network (CNN) model for gesture classification tasks across various arm postures. CWT is a time-frequency domain feature that has shown promising results in MMG hand gesture recognition literature [29], along with other advanced feature sets such as the Fusion of Time Domain Descriptors (fTDD) or Wavelet Scattering Transform (WST) [30]. CWT can capture gradual changes in both temporal and frequency domains, which makes it suitable for convolutional methods and the domain adaptation context. It is worth noting that this CWT feature set, in combination with the CNN model, has not previously been used for domain adaptation techniques in the hand gesture recognition context, including in the extensively studied EMG literature.

III. METHODS

This section describes the hardware, recording protocols, and classification pipeline developed in this work. A summary diagram of the proposed architecture and the conducted experiments is shown in Fig. 1.

A. Datasets

This work uses a gesture dataset recorded from 13 participants (3F/10M, average age: 28.9 ± 3.4 years old) performing hand gestures in different arm postures. Each participant was given a detailed illustration of the experimental protocols and gave his/her informed consent. This study received ethical approval from the Imperial College Research Ethics Committee (ICREC), U.K. (ICREC Reference number: 15IC3068).

B. MMG Recording

An armband consisting of six microphone-based MMG sensors (Fig. 2) was placed on the dominant arm at the largest

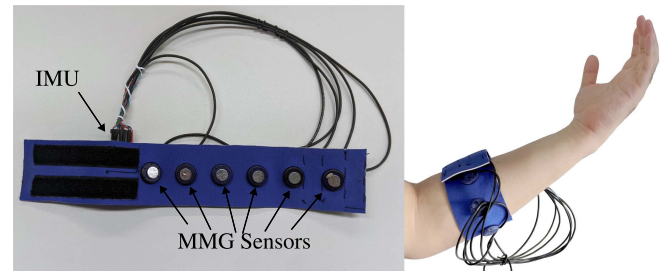


Fig. 2. Armband equipped with six microphone-based MMG sensors and an IMU.

TABLE I
DESCRIPTION OF THE SELECTED ARM POSTURES

Arm Posture	Description
1	Shoulder relaxed, elbow relaxed, neutral wrist
2	Shoulder flexion 45°, elbow relaxed, neutral wrist
3	Shoulder flexion 90°, elbow relaxed, neutral wrist
4	Shoulder flexion 135°, elbow relaxed, neutral wrist
5	Shoulder relaxed, elbow flexion 45°, neutral wrist
6	Shoulder relaxed, elbow flexion 90°, neutral wrist
7	Shoulder relaxed, elbow flexion 135°, neutral wrist
8	Shoulder flexion 90°, elbow flexion 90°, neutral wrist
9	Shoulder flexion 90°, elbow relaxed, pronated wrist
10	Shoulder flexion 90°, elbow relaxed, supinated wrist

section of the forearm. Each sensor was evenly spaced (4 cm apart) on the arm to approximately lie on the Pronator Teres, Flexor Carpi Radialis, Flexor Carpi Ulnaris, Extensor Digitorum Communis, Extensor Carpi Radialis Longus, and Extensor Carpi Ulnaris muscle, respectively. These muscle sites have shown discriminatory information with MMG signals when performing hand gestures [10], [31]. Additionally, an Inertial Measurement Unit (IMU) was attached to the armband to acquire arm orientation during the experiments and determine the static state of the arm before recording the gesture contractions. This method ensured that the motion-induced artifacts were not presented in the recorded samples [20]. The MMG sensor and IMU were developed in-house with the details presented in Woodward et al. [23] and Wilson et al. [4] works.

C. Recording Protocol

Gestures were recorded from 10 arm postures as described in Table I and illustrated in Fig. 3. This set of arm postures

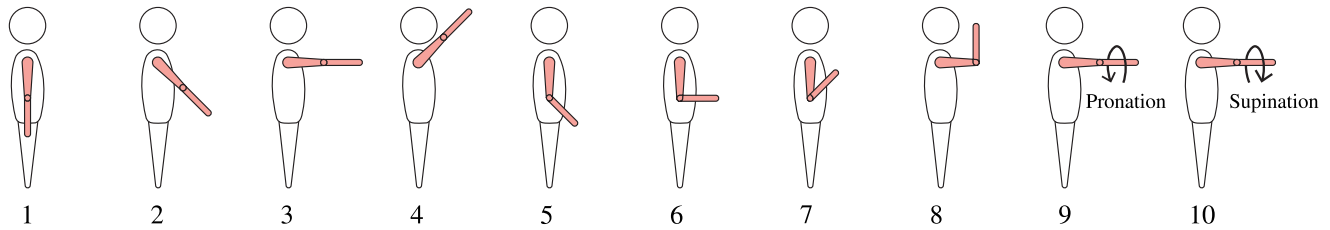


Fig. 3. Selected arm postures. Postures 1-4 cover a range of shoulder motions; Postures 5-8 cover elbow angle movements; Postures 9-10 require 90° wrist pronation & supination.

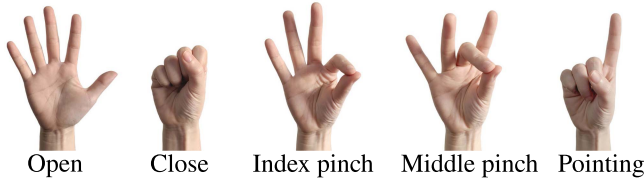


Fig. 4. Selected hand gestures.

was selected from the frequently used postures in hand gesture recognition research [18]. All arm postures lie on the sagittal plane, which helps simplify the movements for subjects to follow and allows isolating the effect of joint movements onto a single movement plane. The postures cover important limb movements such as elbow, shoulder, and wrist rotation, which have been shown to significantly affect muscle activities [12], [32].

Since a single forearm IMU may not accurately capture wrist rotation movement, a manual positioning method is used to control wrist position during pronation and supination postures (postures 9 and 10). The wrist is kept horizontally on the volar and dorsal sides, rotating approximately 90° from a neutral position used in other postures. The wrist position was visually inspected by helper personnel before each recording session to confirm it was within the desired range and ensure consistency of the postures.

Participants were instructed to perform five hand gestures (open hand, close hand, index finger pinch, middle finger pinch, and pointing) as shown in Fig. 4. Each gesture was performed five times consecutively in each arm posture, while the participant was instructed to rest adequately before moving to a new arm posture. The recording was repeated four times, with a total sample size of 1,000 examples for each participant (20 examples $\times$ 5 gestures $\times$ 10 arm postures).

D. Data Processing

MMG signals were sampled at 1 kHz and processed with an Infinite Impulse Response (IIR) 2nd-order Butterworth filter between 1 to 100 Hz. This range of frequencies has been shown to provide significant discriminatory information for forearm MMG [33], [34]. During experiments, hand gesture events were automatically detected by comparing the root sum square (RSS) value of all sensors' outputs with a threshold value, which was adjusted for each individual before the recordings. After detecting the hand gesture events, a segment of 200 ms was extracted from the beginning part of the contraction. This segmentation

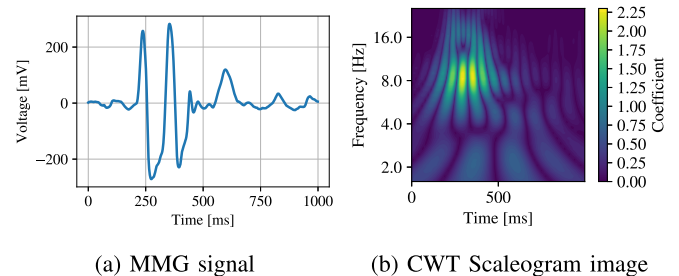


Fig. 5. Example of CWT features extracted from MMG signals

size was selected to ensure a balance between adequate response time and sufficient discriminatory information from the gestures [35], [36].

The MMG segments from all sensors were converted with Continuous Wavelet Transform (CWT) into scaleogram images (Fig. 5) before being forwarded to the classifiers. In this work, the Morlet wavelet was used in CWT conversion along with the frequency scales of eight voices per octave. This number of voices per octave, which translates to the frequency band resolution, was selected to provide sufficient output detail while keeping feasible computational costs for CWT to be used in real-time applications.

E. CNN & DACNN Architecture

A 3-layer Convolutional Neural Network (CNN) ending with a global average pooling layer was developed to classify CWT feature images and provide gesture prediction. This architecture is inspired by the similar domain-adversarial training works (DANN [25] and SCADANN [27]) mentioned in Section II. The structure of convolutional kernels and layers (Fig. 6) was adjusted for CWT features and the corresponding range of MMG frequencies while aiming to achieve short training and response time for the model. The CNN model consists of 36,727 learnable parameters and employs Adam optimizer with a batch size of 32. Learning rate scheduler was applied during training to reduce the learning rate ($lr = 0.005$) by a factor of 0.5 with the patience of 10 epochs. The model was trained for 150 epochs, which showed no significant improvement in loss optimization afterward.

For unsupervised domain adaptation, a domain classification head was added to the model in parallel to the class classification head, as shown in Fig. 6. For simplicity, this new model is called Domain-Adversarial Convolutional Neural Network (DACNN) in this work. While labeled samples from the source domain

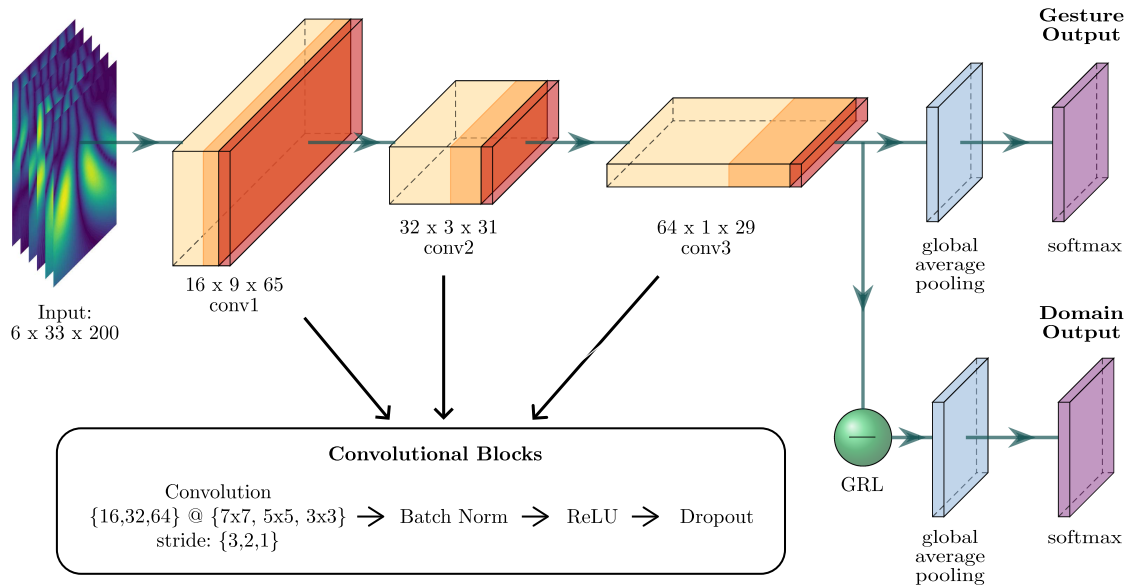


Fig. 6. The DACNN structure with 3 convolutional blocks each consists of a convolutional layer, batch normalization, ReLU, and dropout layer. The class classifier and domain classifier head, which consisted of a global average pooling and softmax layer, were attached to the end of the model. Gradient Reversal Layer (GRL) was implemented before the domain classifier head for domain-adversarial training purposes.

were used for gesture classification training similar to the CNN model, unlabeled samples from the target domain were added to the training process for the model to classify the domain of each sample. A Gradient Reversal Layer (GRL) was implemented in the domain classification head to invert the loss during optimization steps. This GRL layer would affect the model to optimize for gesture classes while simultaneously making itself unable to tell the difference between domains [25].

With this implementation, the DACNN model would learn to generalize class prediction from the source domain while focusing on indiscriminate features between domains. This leverage use of unlabeled samples from the target domain would create a model with better generalization ability on a target domain compared to supervised training with only source domain samples.

IV. EXPERIMENTS & EVALUATIONS

A. Multiple Arm Postures Classification Experiments

The experiments were conducted to explore the muscle activity relationships among arm postures and to show the improvement of the proposed domain adaptation method compared to conventional supervised training methods. The classifiers were trained and tested with samples from combinations of the 10 arm postures. In each training and test scenario, 5-fold cross-validation was used to get the average prediction accuracy of each classifier as the performance metric. The results can be categorized into intra-posture (train and test with the same posture) and inter-posture (train with one posture and test with another). The datasets from each subject were used separately in the experiments, with the average values among subjects reported as the results.

Initially, a CNN model with identical layers to the DACNN model but without the domain classification head was used

for supervised training on the datasets. This is to show the capabilities of CNN and CWT features to extract information from MMG signals, along with the muscle relationships between arm postures. Afterward, unsupervised learning was performed with all arm posture cases using the proposed domain adaptation method (DACNN). Samples from training postures were used in supervised learning of the model, while unlabeled samples from test postures were provided for unsupervised domain adaptation. The model then progresses to adapt itself from the supervised trained posture to the test posture by reducing its ability to predict the domain of the received datasets.

With the above workflow, two tests were conducted in this work using the multiple arm postures dataset.

1) Classifiers Comparison: A linear SVM and CNN model were trained with the same datasets to provide comparison results to the proposed DACNN model and to demonstrate the result of the domain adaptation method. SVM is a classic and popular classification technique that has achieved great results in hand gesture recognition research. Using SVM with time-domain features has been shown to provide good accuracy for MMG signals in controlled environments throughout MMG gesture recognition literature [10], [20]. Ten time-domain features that were shown to achieve significant discriminatory information from MMG signals for hand gesture recognition were selected and used with the SVM model [10].

2) Segment Size Variation: In this work, CWT features were extracted from 200 ms segment of gesture contraction and used with the classifiers. This segment size was selected according to the recommended response time for human-machine interface applications to be adequately reactive for users [35], [36]. However, from further inspection on the recordings, the length of gesture contractions was shown to range from ≈ 600 to 1000 ms. A large portion of information could potentially exist in the gestures and still not fully utilized by the proposed technique.

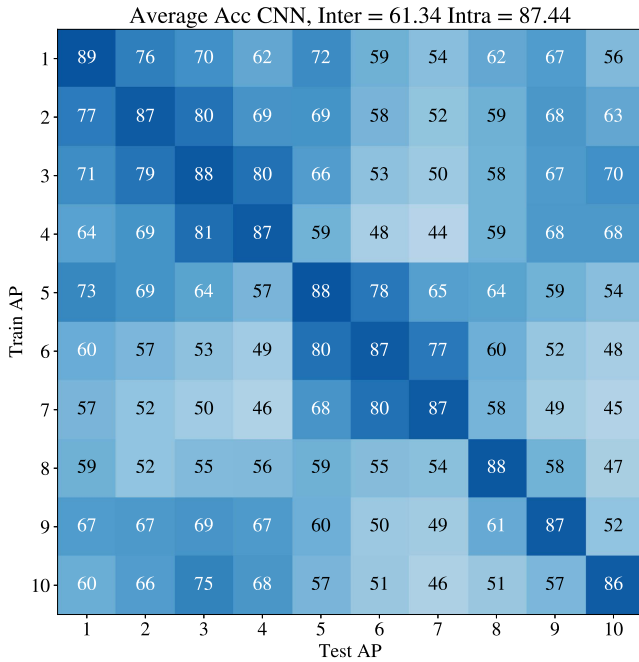


Fig. 7. Prediction accuracy in different training and test arm postures (AP) averaged from all subjects using CNN classifier and CWT features.

For this reason, different segment sizes up to 600 ms with 50 ms step size were extracted and tested with the DACNN model to explore the relevant information in the gestures, which could help further improve the generalization of the model.

B. Results

1) *Arm Posture Relationships*: The average prediction accuracies of the CNN model (without domain adaptation technique and domain classification head) when trained and tested with different arm postures, averaged among all subjects, are shown in Fig. 7. The model could achieve the average accuracy of 87.44% in intra-posture scenarios (diagonal values), while the model accuracies in inter-posture scenarios (non-diagonal values) are significantly lower, with the average value of 61.34%.

From Fig. 7, some arm postures were shown to have similar discriminatory information to other postures. For example, the inter-posture accuracy (non-diagonal values) gradually dropped when the arm posture transitioned from Posture 1 to Posture 4 (Fig. 8(a)). These postures involve shoulder rotation from a relaxed posture while maintaining a fully extended elbow angle. Additionally, a similar decrease in inter-posture accuracy was shown when the arm posture moved from Posture 1 to 5, 6, and 7 (Fig. 8(b)), representing flexion of the elbow while keeping the shoulder relaxed (upper arm at a vertical position). Lastly, wrist rotation also showed a similar effect with gradually lower accuracy when the wrist was supinated or pronated further from the training posture (Fig. 8(c)).

These characteristics indicate the patterns of muscle activities during gesture contraction that exist within the range of arm postures, which MMG sensors could detect. The information changes among arm postures could be discerned with the

proposed CNN model and CWT features, especially in postures with similar joint angles. Furthermore, these gradual information changes could benefit from the domain adaptation technique, which aims to adapt the model by looking at the similarity between domains.

2) *Classifiers Comparison*: The prediction accuracies of the DACNN model, which applied unsupervised training to the test datasets, are shown in Fig. 9. Compared with CNN results (Fig. 7), DACNN can provide a similar range of prediction accuracy across the combinations of train and test arm postures. A noticeable difference is that multiple inter-posture results of DACNN were shown to have small improvements, whereas the intra-posture results are identical at a glance.

Additionally, to compare the classifier results with a simpler metric, the accuracies of each classifier from all subjects are shown as box plots in Fig. 10, along with mean values. The average inter-posture accuracy of the SVM, CNN, and DACNN models were 58.81%, 61.34%, and 64.29%, respectively. On the other hand, the average intra-posture accuracy of each model was 84.37%, 87.44%, and 87.43%, respectively.

To determine the statistical difference between classifier results, Wilcoxon signed-rank test [37] was used by pairwise testing on cross-validation accuracy of each combination of training and test arm postures. The classifier accuracy difference was determined to be statistically significant with ($p < 0.01$). The results show that the CNN classifier with CWT features could improve accuracy compared to SVM by 2.53% for inter-posture results ($p < 0.001$). In addition, implementing domain adaptation (DACNN) could further improve the inter-posture accuracy by 2.95% compared to CNN ($p < 0.001$).

For intra-posture results, the average accuracy of the CNN model was improved by 3.07% compared to SVM ($p < 0.001$). On the contrary, there are no significant differences between CNN and DACNN intra-posture results ($p = 0.804$). This is because the domain adaptation technique receives no benefit when performing unsupervised learning within the same arm posture in intra-posture cases.

Although the interquartile ranges of the inter-posture accuracies shown in the left picture of Fig. 10 are relatively small ($< 22.5\%$) for all models (i.e., SVM, CNN, and DACNN), the ranges between the maximum and minimum accuracies across all subjects are quite large. This result highlights the challenge of hand gesture recognition tasks across different arm postures, which is the main issue this paper aims to address. It can be seen that the box plots in the left picture of Fig. 10 are narrower with the proposed model compared to other non-domain adaptation models, indicating the DACNN can reduce the variance in inter-posture accuracies across subjects by extracting domain invariant information that exists across different arm postures.

It is worth noting that implementing a stricter recording protocol to normalize the subject's movements or implementing a more intensive muscle identification protocol could further help improve the consistency of the information across subjects. However, these aspects required a trade-off with the practicality of the HGR system. The recording protocol in this paper was selected for users to be able to perform gestures naturally and without intensive preparation between each usage.

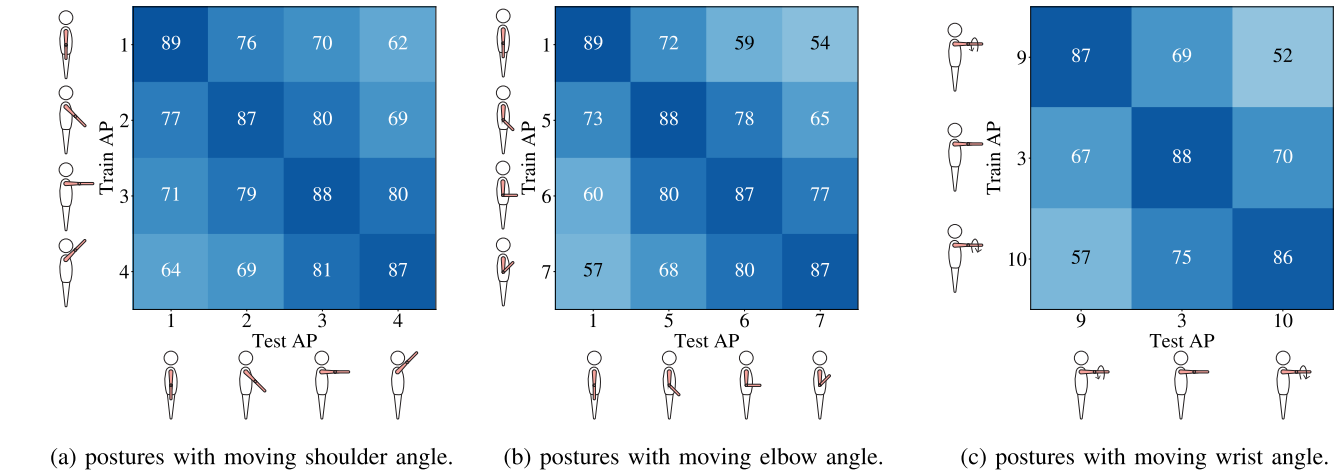


Fig. 8. CNN prediction accuracy on combinations of arm postures.

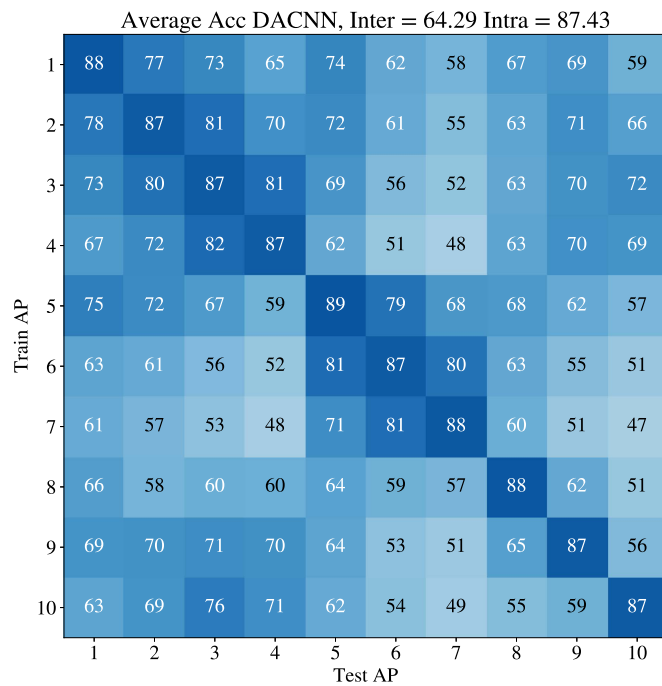


Fig. 9. Prediction accuracy in different training and test arm postures (AP) averaged from all subjects using DACNN classifier.

3) Segment Size Variation: The inter-posture and intra-posture accuracies of the DACNN model with different segment sizes are shown in Fig. 11. The results show an increasing trend with larger segment sizes in both cases. For inter-posture cases, the average accuracy increased significantly with each increment and reached a plateau after around 350 ms. The average inter-posture accuracy was 42.20% at 50 ms, which increased to 68.91% at 350 ms and 71.75% at 600 ms. For intra-posture cases, the average accuracy increased significantly with each increment and reached a plateau after around 400 ms. The average intra-posture accuracy was 56.61% at 50 ms, which increased to 91.43% at 400 ms and 92.32% at 600 ms.

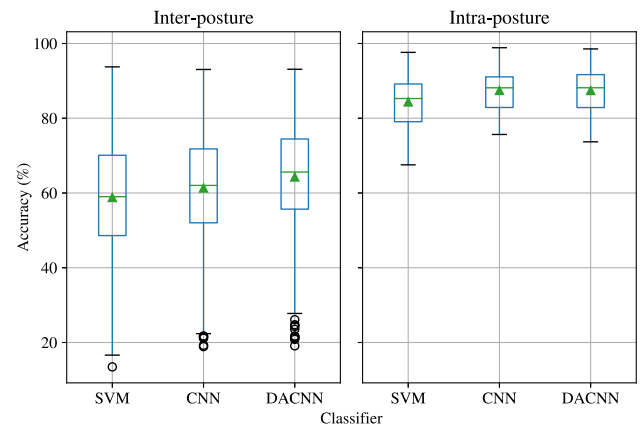


Fig. 10. Inter/Intra-posture accuracies from all arm postures, comparing between classifiers. Green triangles indicate the average value.

C. Discussion

1) Arm Posture Relationships: The results demonstrate that the proposed CNN architecture with CWT features can achieve consistent intra-posture accuracy across arm postures. The CNN model's intra-posture accuracies were higher than 86% in all arm postures (Fig. 7). However, the task of classifying hand gestures on arm postures deviated from the training postures was shown to be challenging. They can have varying results depending on the range of motion related to the training posture. For example, the inter-posture accuracy can drop significantly to 44% when the model was trained and tested with arm postures 4 and 7, respectively. This pair of postures has a large difference in joint angles (one with 135° elbow flexion and another with 135° shoulder flexion), which could cause muscle activities to be entirely different.

2) Classifiers Comparison: The DACNN model demonstrated improved generalization compared to other supervised learning classifiers, as shown in Figs. 9 and 10. The results indicate that the domain adaptation technique can effectively improve inter-posture accuracy across all subjects and most

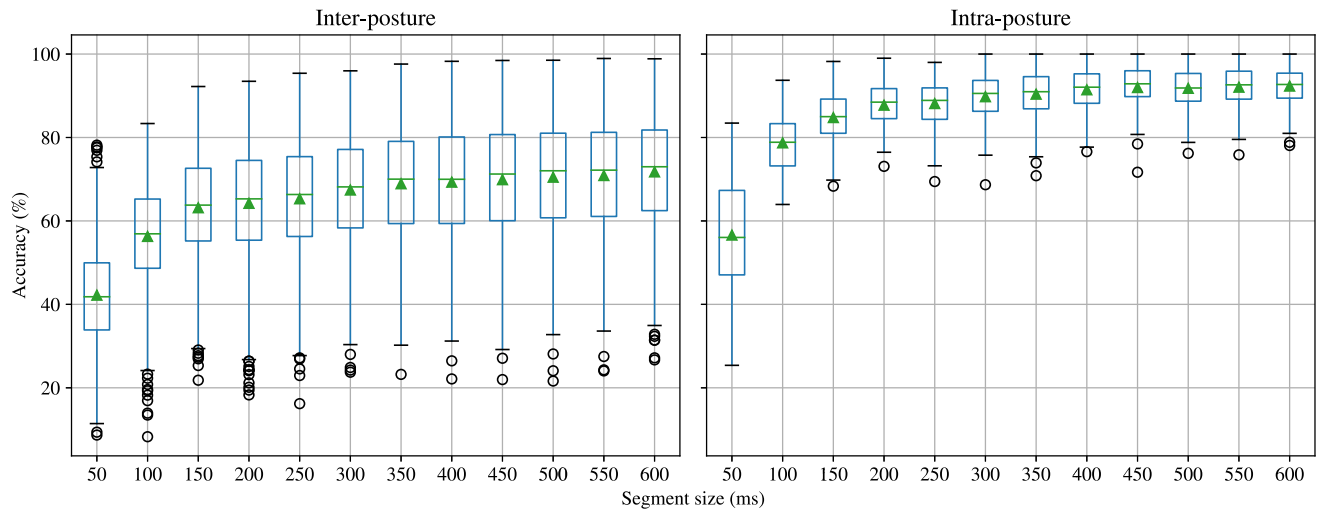


Fig. 11. Average accuracies of DACNN classifier using different segment sizes.

combinations of arm postures. Using the 10 arm postures selected from popular postures in research, the proposed DACNN shows that it can pick up discriminatory information from muscle activities across arm movements. This reflects the potential for usage in non-laboratory conditions where users could perform gestures in different arm postures during usage.

It is worth noting that the experiments were focused on training with a single posture to highlight the significant difference in information across different arm postures. The classifiers would perform better if samples from multiple arm postures were provided for training. However, the training time required would significantly increase with more recording postures. This trade-off between the number of data and preparation time is not always clear to decide and should depend on the specific requirements and goals of each hand gesture recognition application.

3) Segment Size Variation: The results show that the DACNN model can extract more discriminative information from gestures with larger segment sizes before achieving a diminished improvement after a specific size. The model performed well with datasets from the same arm posture even with small segment sizes, achieving average intra-posture accuracies higher than 78% with a small segment size of 100 ms, as shown in Fig. 11. Conversely, the model's accuracy decreased significantly when dealing with datasets from different arm postures (inter-posture). Nevertheless, the model could gradually achieve better accuracy with larger segment sizes, indicating the utilization of underlying information in the MMG signals.

With this result, a segment size of 200 ms was determined to be adequate, achieving good accuracy (87.43% in intra-posture and 64.29% in inter-posture cases) while still maintaining a reactive response time for the application. The investigation showed that using a larger segment size, such as 600 ms, could further improve inter-posture accuracy, reaching up to 71.75%. Although larger segment sizes may impact prediction time, this knowledge could be leveraged in applications where response time is less critical in order to achieve a higher generalization level.

V. CONCLUSION

This study proposed a new hand gesture recognition pipeline and its training paradigm tailored for MMG activities from forearm muscles. The study focused on addressing the problem of arm posture variability, which is a major challenge in non-laboratory environments. A new combination of CWT features and a CNN classifier were used to extract discriminatory information from hand gesture contractions. In addition, a domain-adversarial training method was implemented to perform unsupervised learning on unlabeled samples from new arm postures to increase generalization ability.

With 10 arm postures, the proposed DACNN classifier can achieve improved inter-posture accuracy compared to CNN and SVM classifiers, achieving an accuracy of 64.29% compared to 61.34% and 58.81%, respectively. In addition, the domain adaptation does not cause any decrease in intra-posture accuracy, with the DACNN classifier achieving average intra-posture accuracy of 87.43%, compared to 87.44% for CNN and 84.37% for SVM classifiers. It is worth noting that the above results were obtained using 200 ms segment of MMG signals, which allows the application to maintain real-time prediction capability.

The results also demonstrated significant muscle activity changes across different arm postures during hand gesture contraction. The changes were shown to be correlated with joint movements, with a gradual decrease in prediction accuracy when the shoulder, elbow, and wrist joints rotated further from the training posture. The findings reflect the capability of the proposed architecture to focus on important features present among different arm postures, enabling improved gesture recognition in non-laboratory environments where users could perform gestures in arbitrary arm postures after initial training in specific positions.

In addition to using the default 200 ms segment size, a further experiment was conducted to explore the discriminatory MMG information that existed in the later parts of gesture contractions. The results showed this information could be effectively utilized by the proposed technique. The classifier achieved

significant accuracy improvement with larger segment sizes, with inter-posture accuracy of 71.75% and intra-posture accuracy of 92.32% using a 600 ms segment size. This finding can be leveraged for applications that allow longer response times or prioritize classifier accuracy.

Although the analysis in this work was conducted offline with the recorded datasets, real-time usage capability is a key design criterion considered throughout the architecture's development. With a regular personal laptop (CPU: Intel i7-8550U) and PyTorch implementation, the model requires less than 40 ms for CWT feature extraction and 2 ms for gesture prediction. The training protocol requires less than 5 minutes to record data for one arm posture from a subject (5 gestures $\times$ 20 repetitions), and the model can train to achieve optimal loss with the dataset in less than 5 seconds. Additionally, an exploratory analysis was conducted to compare these timing aspects of CWT with other advanced MMG features from the literature, with the results showing that other features can take significantly longer time while providing a similar level of discriminatory information (e.g., WST feature extraction can take ≈ 120 ms with similar wavelet parameters, while the CWT feature extraction requires around 40 ms). This highlights the capability of the selected CWT features to deliver highly discriminatory information across various arm postures, support real-time applications, and be suitable for domain adaptation techniques.

This study demonstrated the developed technique's ability to adapt through different datasets from unseen domains without the need for labeled samples. While the arm posture problem was used in this investigation, the technique could be used to address other non-laboratory problems that exhibit gradual changes in muscle activities between sessions, such as fatigue or sensor shifts. The domain-adversarial training technique has advantages in teaching classifiers to focus on domain-invariant features, regardless of the type of changes. According to the results of our previous work [10], adding a small size of data from the target domain to the model training process can improve the accuracy of hand gesture recognition for the target domain. The proposed DACNN follows a similar principle but uses only unlabelled samples from the target domain. Furthermore, the training scheme only requires an auxiliary domain classification layer and a gradient reversal layer to be implemented alongside the main model. This flexible implementation could be easily applied to many other learning models, thus providing great potential for exploring and improving physiological studies that aim to address the dynamic factors in real-world applications.

For limitations and future works, potential improvements for this research include implementing techniques to extract more discriminatory information from MMG signals while maintaining an adequate prediction time. Data augmentation could be implemented during the training phase to create a more robust model with fewer segmentation errors. Alternatively, fusing inertial information with MMG for better gesture prediction could be beneficial. In this research, a single IMU was used to confirm forearm orientation for domain adaptation analysis, i.e., ensuring all subjects performed the required arm postures correctly. It should be noted that no IMU data was used in the machine learning pipeline (i.e., SVM, CNN, and DACNN)

for gesture recognition. However, a full practical setup would require a comprehensive limb position identification system, including both the forearm and upper arm. For continuous arm position tracking, the system has to accurately identify arm posture and assign the domain labels in real-time, as well as be able to determine the position in 3D space.

For practical application, hand gestures may change during complex 3D arm movements, causing different changes in the relevant muscle activities. Although the DACNN can extract domain-invariant information to improve the recognition accuracy for a new 3D arm posture, the inter-posture accuracy might be different from the experimental results due to the different muscle characteristics. According to the results shown in Figs. 7–9, a high inter-posture recognition accuracy can be guaranteed when using the samples from some specific relevant arm postures. It is worth further conducting rigorous experiments to explore the relationships among different arm postures in future work, which could help build a dataset collected with a limited number of hand gestures for accurately recognizing gestures in complex 3D movements. Our team has patented the myographic sensing system implemented in this investigation, which is capable of providing real-time data with high accuracy for practical, real-world applications. Translational efforts addressing larger-scale repeatability and data collection at scale will be targeted in future work.

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